Toronto Airbnb Listings

Modelling Price Determinants and Testing for Structural Change Under Municipal Short-Term Rental Licensing

An Application of Multiple Regression, Regression Diagnostics, and Hypothesis Testing

MMA 860 — The Acquisition and Management of Data
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1 Executive Summary

Every host who lists a property on Airbnb in Toronto must decide what nightly price to charge, typically without a systematic way to benchmark that price against comparable listings. Toronto's short-term rental (STR) market is large and heterogeneous. This analysis draws on 15,869 active listings scraped from Inside Airbnb between May and June 2026, spanning 140 neighbourhoods and four room types, with nightly prices ranging from $6 to over $11,000 and a mean of $275 against a median of $193. Since 2020, short-term rentals in Toronto have been governed by Municipal Code Chapter 547, which requires operators to register with the city and display a valid registration number. In this dataset, 35.3% of listings display no such number. This report uses multiple regression to identify the determinants of nightly price and a series of diagnostic and hypothesis tests, including a Chow test, to assess whether licensed and unlicensed listings represent one pricing market or two structurally distinct ones.

We built a hedonic pricing model in two stages. Stage 1 estimated a baseline regression of price on accommodation size, room type, neighbourhood, host characteristics, and licensing status; a log-price refinement fit the data dramatically better (R^2 rose from 0.30 to 0.73) and is the model used for all subsequent inference. Stage 2 subjected that model to the diagnostic and hypothesis-testing process required to trust it. Variance Inflation Factors (VIF) confirmed no problematic collinearity; a Breusch-Pagan test confirmed heteroskedasticity, addressed with heteroskedasticity-consistent standard errors; Cook's Distance identified roughly 5% of listings as unusually influential, and excluding them left the headline findings unchanged; a joint F-test confirmed that neighbourhood matters as a block; and a Chow test decisively rejected the hypothesis of one pooled pricing relationship for licensed and unlicensed listings ($F = 62.07$, $p \approx 0$).

Licensed listings average $331 per night versus $165 for unlicensed listings, roughly double, and the Chow test shows this is not merely a level shift but a structurally different pricing relationship. A follow-up test of whether professionalization signals (host portfolio size, identity verification, amenity count) explain this gap found they account for only about 13% of the licensing coefficient's effect; most of the licensing premium remains unexplained by the variables available in this dataset. A separate test linking each listing's pricing residual to its calendar-derived occupancy rate found a small but statistically significant negative relationship. This may mean listings priced above what the model predicts for their characteristics see measurably lower occupancy, providing modest empirical support for the overpricing risk that motivated this analysis.

We recommend hosts price primarily off accommodation size, room type, and location; treat licensing status as a genuine structural segment rather than a simple additive adjustment; and consider a large positive pricing residual as an early warning that a listing may be priced above what comparable properties in its segment command. For the City, the persistence of a large, largely unexplained licensing premium, combined with the finding that unlicensed listings skew toward *larger* host portfolios rather than smaller ones, is a useful and somewhat counter-intuitive data point for prioritizing Chapter 547 enforcement.

2 Introduction and Problem Statement

Every host who lists a property on Airbnb in Toronto must decide what nightly price to charge, typically without a systematic way to benchmark that price against comparable listings. Toronto's short-term rental market is large and heterogeneous. This analysis draws on 15,869 active listings scraped from Inside Airbnb between May and June 2026, spanning 140 neighbourhoods and four room types, with nightly prices ranging from $6 to over $11,000 and a mean of $275 against a median of $193. This is a right-skewed distribution typical of accommodation pricing (Figure 1).

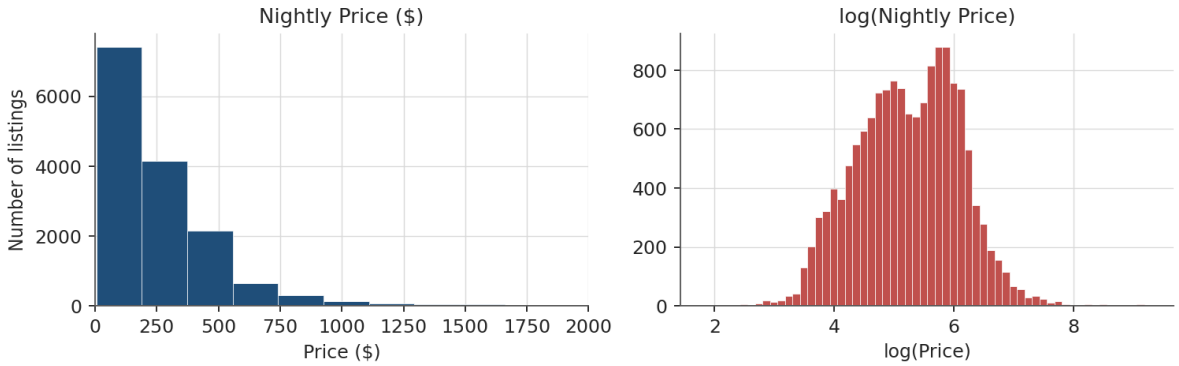


Figure 1. Distribution of Toronto Airbnb nightly price, raw and log transformed.

Two related pricing problems follow from this heterogeneity. Hosts who price below what comparable listings command leave revenue on the table, a cost that compounds over hundreds of booked nights per year. Hosts who price above the market for their listing's characteristics risk suppressed occupancy relative to competitively priced comparable units.

A second, related question concerns Toronto's regulatory environment. Since 2020, short-term rentals have been governed by Municipal Code Chapter 547, which requires operators to register with the City and display a valid registration number, restricts entire-unit rentals to 180 nights per calendar year, requires the property be the operator's principal residence, and applies an 8.5% Municipal Accommodation Tax on stays under 28 nights. In this dataset, 35.3% of listings display no registration number. This raises a natural question: does the price-determination process differ between listings that display a registration number and those that do not? That is, does regulatory compliance status coincide with a structurally different pricing regime?

This report addresses both questions using multiple regression to identify the determinants of nightly price, and a battery of diagnostic and hypothesis tests, including a Chow test, to assess whether that pricing relationship is stable across the licensed and unlicensed segments of the market.

3 Data

3.1 Data Sources

Data Element	Source	Notes
Toronto Airbnb listings (detailed)	Inside Airbnb: data.insideairbnb.com/canada/on/toronto	15,869 listings, 90 fields
Toronto Airbnb calendar	Inside Airbnb	8.11 million date-availability rows (listing_id, date, available, min/max nights); used for the occupancy
Short-term rental licensing rules	City of Toronto, Municipal Code Chapter 547	Used to interpret the license field and define the licensed dummy variable

3.2 Data Cleaning and Preparation

Price was stored as text in the dataset (e.g. "$99.85") and was converted to numeric.

Below table summarizes missingness in the fields considered for modeling and how each was treated; full detail appears in Appendix D.

Field	% Missing	Treatment
neighbourhood_group_cleansed	100.0%	Dropped (fully empty for Toronto)
instant_bookable	100.0%	Dropped (fully empty in this snapshot)
bathrooms (numeric)	54.0%	Superseded by parsed bathrooms_text (0.2% missing)
beds	52.4%	Imputed: property-type median
license	35.3%	Not missing data. Defines the unlicensed group
bedrooms	25.7%	Imputed: property-type median
review_scores_rating	17.9%	Imputed: sample median + missingness flag retained
price	5.4%	Rows dropped (dependent variable)
minimum_nights	<0.1%	Rows dropped (5 listings)

Table 1. Missing-data audit and treatment.

Rows missing price (861, 5.4%) & minimum_nights (5) were dropped, since neither the dependent variable nor a near-universal field can be sensibly imputed for modeling purposes. The modeling dataset stands at 15,003 listings after cleaning. The raw license field was converted into a binary licensed indicator: 66.1% of listings display a licence number. The 140 raw neighbourhoods were consolidated into the 14 most common categories plus an "Other" bucket for the regression's dummy-variable block, since using all 140 individually would leave many neighbourhood coefficients estimated on a handful of listings.

3.3 Feature Engineering

Room type and the grouped neighbourhood variable were one-hot encoded, each dropping one category as the regression baseline (Entire home/apt for room type; the Annex for neighbourhood). host_is_superhost was converted to a 0/1 flag. The cleaned dataset (15,003 listings, 26 predictor columns after encoding) was split 80/20 into a training set (12,003 listings), used for model fitting and all diagnostics and hypothesis tests, and a held-out test set (3,000 listings), used only for the out-of-sample validation in Section 6 of the script.

4 Model 1: Multiple Regression Price Model

4.1 Model Specification

The regression model treats nightly price as a function of listing characteristics (size, capacity, room type), host characteristics (superhost status, tenure, review score), and location (neighbourhood), plus the licensing indicator at the centre of this analysis — a standard hedonic pricing model:

$$Price = \beta 0 + \beta 1 \cdot Accommodates + \beta 2 \cdot Bedrooms + \beta 3 \cdot Beds + \beta 4 \cdot Bathrooms + \beta 5 \cdot RoomTypej + \beta 6 \cdot Neighbourhoodk + \beta 7 \cdot Superhost + \beta 8 \cdot HostTenure + \beta 9 \cdot ReviewScore + \beta 10 \cdot MinNights + \beta 11 \cdot Licensed + \varepsilon$$

4.2 Collinearity Diagnostics (VIF)

accommodates, bedrooms, beds, and bathrooms are correlated with each other in the 0.6-0.9 range. Here, raw pairwise correlation is not the right diagnostic for deciding whether to drop a variable, since two variables can be highly correlated with each other, yet both still contribute cleanly once the rest of the model is accounted for. Variance Inflation Factor (VIF) has been used to do the checking:

Variable	VIF
accommodates	4.00
bedrooms	3.94
beds	2.57
bathrooms	1.99

Table 2. VIF among the four size variables.

Every value is comfortably below the conventional concern threshold of 10, so all four size variables were retained. The full-model VIF (26 predictors) tops out at 8.39 for the "Other" neighbourhood bucket and 5.32 for Waterfront Communities.

4.3 Baseline Results

The model was assessed in following order, the overall F-test first, then individual t-tests, then diagnostic plots, then R^2.

The F-statistic's p-value is effectively 0, so the null that all coefficients are jointly zero is rejected decisively. Most individual predictors are significant at $\alpha=0.05$ level. A few individual neighbourhood dummies are not, which is expected when the omitted baseline (the Annex) is itself a moderately priced, well-known area, and is tested as a block in Section 4.3.

Statistic	Value
R^2	0.303
Adjusted R^2	0.301
F-statistic	199.8 ($p \approx 0$)
Observations (training set)	12,003
Predictors	26
AIC	170,178

Table 3. Baseline (raw-price) model fit statistics. Full coefficient table in Appendix A.

Each additional bathroom is associated with the largest single price jump among the size variables (+$105/night, holding other characteristics fixed); a private room is associated with a substantially lower price than the omitted base case, entire home/apt (−$64/night); and licensed carries a large, positive, significant coefficient (+$77/night).

4.4 Refined Model: Log-Price Transformation

The raw-price distribution is strongly right-skewed (mean $275 vs. median $193, maximum $11,042), and the residual diagnostics in Section 4.1 confirm this produces non-constant error variance. Re-estimating with log (Price) as the dependent variable fits dramatically better. R^2 rises from 0.303 to 0.730 and AIC drops from 170,178 to 14,962. This is a strong signal that the raw-price model was mis-specified, not merely heteroskedastic, and that these predictors affect price multiplicatively rather than additively. Each coefficient can now be read as an approximate percentage effect on price via (e^β − 1) × 100%. The log-price model, refit with heteroskedasticity-robust standard errors (Section 4.1), is the model used for every subsequent test in this report; the full coefficient table appears in Appendix A.

Selected results:

- each additional bathroom is associated with roughly +14.0% in price
- switching from a private room to an entire home/apt is associated with a much larger jump (private room: −44.1% relative to the entire-home baseline)
- each additional minimum-night requirement is associated with a small per-night price discount (−1.4%)
- and licensed carries a large, positive, significant effect on price (+37.2%)

'Beds' carries a statistically significant negative coefficient (−5.6%) in the full model, even though it is strongly positively correlated with price on its own (+51.8% in a bivariate fit - Figure 4). This is not a collinearity impact. 'beds' VIF is 2.57, well under the concern threshold established. It is a genuine partial effect. For two listings with the same guest capacity, bedroom count, and bathroom count, the one that packs in more beds is priced lower, consistent with extra beds signalling bunk-style or shared-sleeping configurations rather than added comfort once capacity is already captured by accommodates. The variable was retained rather than dropped. The finding is substantively interesting, and the low VIF confirms it is not destabilizing the rest of the model.

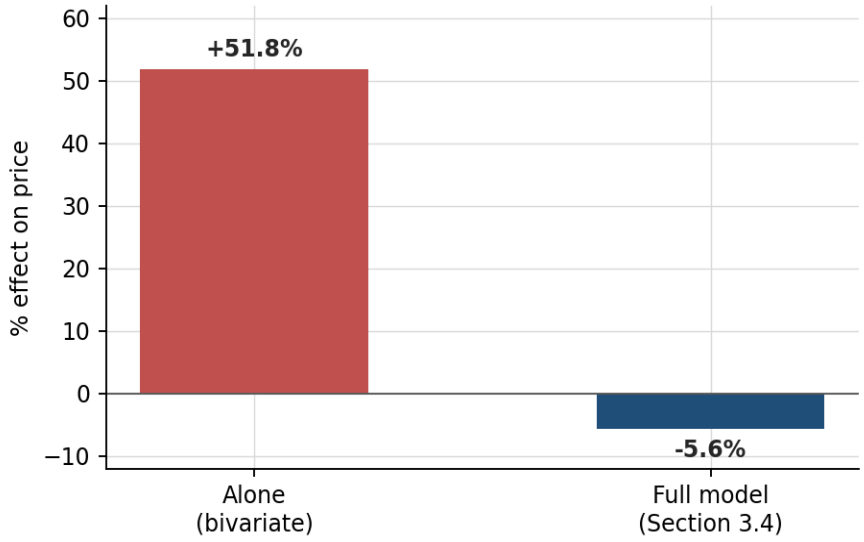


Figure 2. Bed's price effect: alone versus holding capacity, bedrooms, bathrooms, and room type fixed.

5 Model 2: Diagnostics and Hypothesis Testing

5.1 Heteroskedasticity and Robust Inference

A Breusch-Pagan test on the baseline model's residuals rejects homoskedasticity (LM = 59.29, p = 0.0002); the residuals-vs-fitted panel for the log model (Figure 3) shows the same underlying spread pattern. After the log transform, the Breusch-Pagan test still rejects homoskedasticity (LM = 561.84, p ≈ 0). It is a somewhat counter-intuitive finding, since the LM statistic is larger, not smaller. This does not mean the transform failed. It fixed the dominant problem (variance scaling with the level of price, driven by a handful of extreme listings), while some remaining heteroskedasticity tied to categorical segments with few observations (e.g., shared and hotel rooms) is still detectable, and a training set of 12,000 observations gives the test enough power to detect even small remaining patterns.

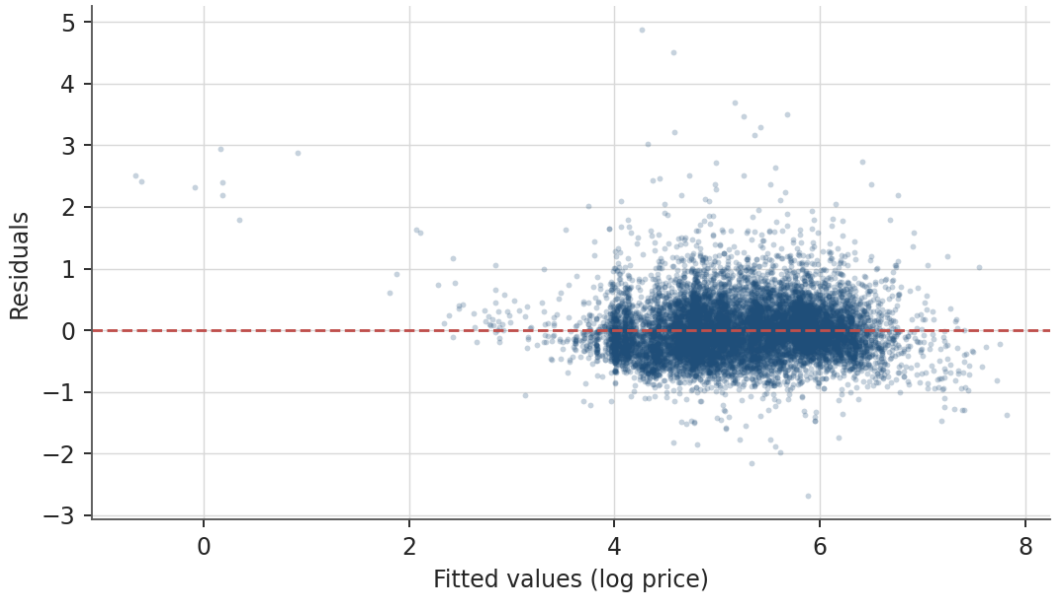


Figure 3. Residuals vs. fitted values, log-price model.

The textbook remedy for this situation is not further transformation, but heteroskedasticity-consistent (HC1) standard errors for inference. Every variable significant under standard OLS remains significant under the HC1-robust version (full comparison, Appendix A).

5.2 Outlier and Influence Diagnostics

Cook's Distance identifies a small share of listings, concentrated among the highest-priced ones, that disproportionately influence the fitted model:

Model	Cook's D threshold (4/n)	Max D	Flagged	% Flagged
Baseline (linear price)	0.000333	0.1229	237	2.0%
Log-price model	0.000333	0.0492	623	5.2%

Table 4. Cook's Distance summary, linear and log-price models.

Re-fitting the log-price model excluding the flagged influential listings improves fit further (R^2 rises from 0.730 to 0.805) and leaves the core size and structural predictors - accommodates, bedrooms, beds, bathrooms, room type, and licensed - stable in sign and within roughly 1-12% of their original magnitude (full comparison, Appendix B). The

headline findings are not a result of a handful of extreme listings; the full sample (all 15,003 listings) was retained for the hypothesis tests that follow.

5.3 Joint F-Test: Does Location Matter?

Several individual neighbourhood dummies in Section 3.4 of the script are not statistically significant on their own. This is expected when the omitted baseline (the Annex) is itself a specific, moderately priced area, and some other neighbourhoods simply do not differ much in price from it. Rather than reading anything into those individual t-tests, the entire block of 14 neighbourhood dummies was tested jointly:

$$H_0: \beta_{nbhd,1} = \beta_{nbhd,2} = \ldots = \beta_{nbhd,14} = 0 \quad vs. \quad H_1: \text{at least one } \beta_{nbhd,k} \neq 0$$

The result is F = 170.90, p ≈ 0.

H_0 is rejected decisively. Location matters jointly, even though several individual neighbourhoods are statistically indistinguishable from the baseline on their own.

5.4 Chow Test: Licensed vs. Unlicensed Pricing

Toronto's Chapter 547 licensing requirement is not a cosmetic label. A licensed, entire-unit registration is capped at 180 nights per year and must be the operator's principal residence. This is meaningfully different economics from an unregistered listing. The Chow test asks whether the *entire* pricing relationship, not just the average price level, differs between the two groups. Every predictor was interacted with the licensed dummy, and a joint Wald test on the licensed main effect plus all 25 interaction terms (26 restrictions) gives F = 62.07, p ≈ 0. H_0 (one pooled pricing relationship for both groups) is rejected decisively.

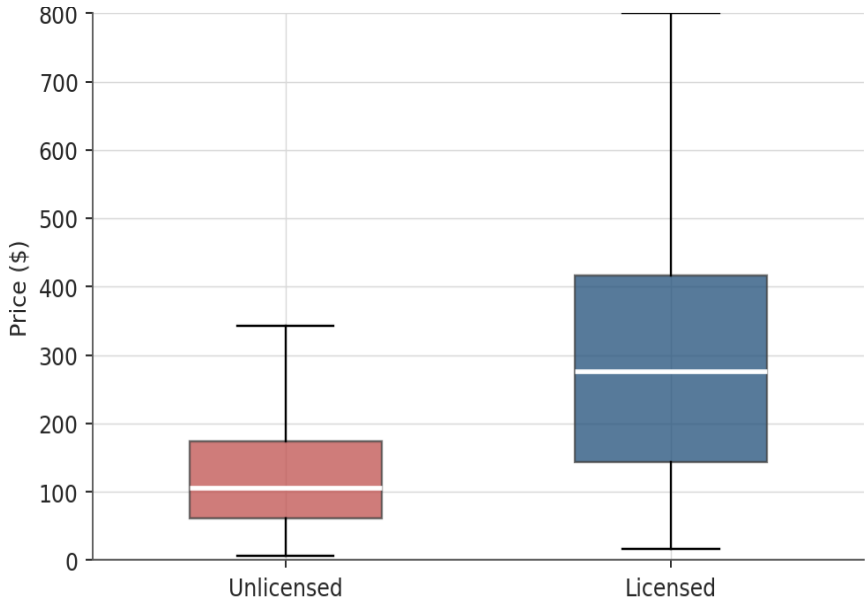


Figure 4. Nightly price: licensed vs. unlicensed listings.

The raw numbers underneath this are striking. Licensed listings average $331/night (median $277, n = 9,920) versus $165/night (median $107, n = 5,083) for unlicensed listings which is roughly double. Because this is a cross-sectional comparison, the Chow test establishes that licensing status coincides *with* a structurally different pricing regime, not that becoming licensed causes a listing's price to change.

6 Extension: Explaining the Licensing Premium and Testing for Overpricing

6.1 Trust and Professionalization Signals

Section 4.4 established that licensed and unlicensed listings have structurally different pricing relationships, and that licensed listings average roughly double the price of unlicensed ones. But a Chow test cannot say 'why'. Three possible rational are worth distinguishing: a compositional effect (licensed listings skew toward pricier room types), a trust/legitimacy effect (a registration number as a signal to guests), and a professionalization effect (licensed operators investing more in furnishing, service, and presentation).

The obvious trust proxies in the dataset host_response_rate, host_response_time, host_acceptance_rate, and host_verifications are 100% missing and could not be used. host_identity_verified, host_listings_count, and a parsed amenities_count were substituted as the closest available signals for the trust and professionalization rational.

A real, somewhat counter-intuitive finding: unlicensed listings belong to hosts with larger portfolios on average than licensed listings.

Group	Mean host_listings_count	Median host_listings_count
Licensed	3.4	1
Unlicensed	14.3	3

Table 5. Host portfolio size, licensed vs. unlicensed listings.

This runs against an intuitive "licensed = professional operator" story, but follows directly from Chapter 547's principal-residence requirement, which structurally excludes large-portfolio operators from licensing most of their units. A large-portfolio operator is, by definition, close to unable to claim all their units as a principal residence. If anything, this suggests licensed leans toward proxying for "small-scale, owner-occupied" rather than "large-scale, professionally managed."

Adding the three rationals to the full-sample log-price model shrinks the licensed coefficient by roughly **13%**:

	licensed coefficient	Implied price effect
Before controls	0.312	+36.6%
After controls	0.270	+31.0%

Table 6. Shrinkage of the licensed coefficient after controlling for trust and professionalization signals.

The added block is jointly significant ($F = 61.73$, $p \approx 0$):

Variable	Coefficient	% effect	p-value
amenities_count	0.001	+0.08%	0.004
host_identity_verified	−0.018	−1.8%	0.549 (not significant)
log(host_listings_count)	−0.050	−4.8%	<0.001

Table 7. Trust and professionalization signals added to the full-sample model.

The shrinkage is driven almost entirely by log(host_listings_count); amenities_count is statistically significant but tiny in practical size; host_identity_verified is not significant at all - unsurprising given 98.4% of hosts in this dataset are already ID-verified, leaving almost no variation for the model to use. Most of the licensing premium survives these controls. It is not purely compositional or purely a trust signal, and something about licensing status itself, or something not captured by these three variables, still matters.

6.2 Linking Price Residuals to Occupancy

Section 1 raised overpricing, not just underpricing, as a cost hosts face: a listing priced above what its characteristics justify should see suppressed occupancy relative to competitively priced comparable units. This was tested directly using "calendar.csv" dataset's 8.11 million date-level availability rows, collapsed into a per-listing occupancy rate.

Merging this onto each training listing's Section 3.4 pricing residual (11,643 of 12,003 training listings had calendar coverage) gives a correlation of −0.069 between residual and occupancy rate. This is small, but with a sample this large, statistically significant ($p < 0.001$). The pattern is easier to read from a quartile breakdown than the raw correlation:

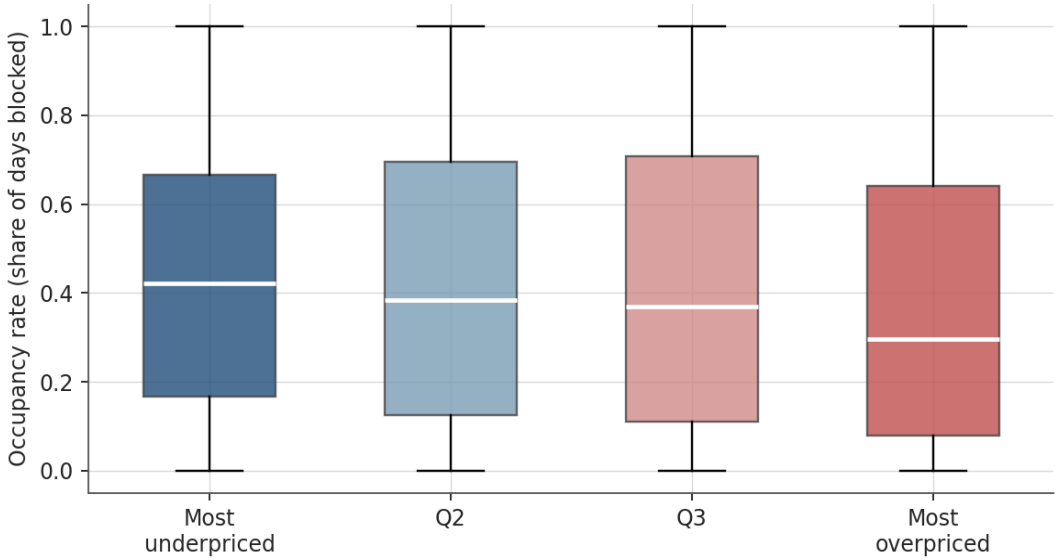


Figure 5. Occupancy rate by price-residual quartile (training set).

Median occupancy falls from 42% for the most-underpriced quartile of listings to 30% for the most-overpriced quartile.

This provides modest empirical support for the overpricing hypothesis raised before. Mispricing on the high side is associated with measurably lower occupancy, not just lower revenue per booking. Though the effect is small next to the structural drivers, and remains correlational, since occupancy also depends on factors outside this model (photos, response time, seasonality).

7 Out-of-Sample Validation

Everything up to this point measures how well the model explains the data it was fit on. The final check is how well it predicts the 20% of listings held out as ‘test in Section 2.3 and never used to fit any model.

Model	Test RMSE ($)	Test MAE ($)
Linear (Model 1)	$357.00	$115.64
Log-linear (Model 1, refined)	$358.15	$98.84

Table 8. Out-of-sample prediction error, linear vs. log-linear model.

RMSE is nearly identical between the two models because RMSE squares errors, it is dominated by the handful of very expensive listings in the test set, where being wrong by a large dollar amount swamps everything else regardless of which model made the prediction. MAE tells a different, more decision-relevant story. The log-linear model's *typical* prediction error is meaningfully lower. For the great majority of hosts pricing a listing anywhere near the $100–400/night range where most of the market actually sits, the log-linear model is the more accurate and more useful one.

8 Recommendations

8.1 For Hosts: Pricing Guidance

- **Size and quality drive price predictably.** Each additional bathroom, bedroom, and unit of guest capacity is associated with a statistically significant price premium. A host adding a bathroom or converting a den to a bedroom has a defensible basis for raising their nightly rate.
- **Adding sleeping capacity only pays off if it comes with more room.** More beds alone, without a matching increase in bedrooms or bathrooms, is associated with a lower price, not a higher one. Packing more guests into the same footprint reads as a budget configuration, not a premium one. A host raising capacity should add a bedroom or bathroom alongside it, not just a mattress.
- **Room type is the single largest lever.** Entire-home listings command a large premium over private rooms; this is the biggest structural decision a host makes, well before fine-tuning.
- **Location matters, but unevenly.** The joint F-test confirms neighbourhood is a real, statistically significant driver of price but not every neighbourhood differs meaningfully from the citywide baseline. Hosts in typical, non-standout neighbourhoods should not expect much of a locational premium or penalty.
- **Overpricing has a measurable occupancy cost.** Listings priced well above what the model predicts for their characteristics see measurably lower occupancy. This is a modest effect next to size, room type, and location, but a real, testable version of the underpricing/overpricing trade-off raised at the proposal of this project.
- **Use the log-linear model, not the raw-price model, for any pricing tool.** It fits dramatically better (R^2 0.73 vs. 0.30) and predicts materially more accurately for typical listings.

8.2 For Policy: The Licensing Finding

The Chow test found that licensed and unlicensed listings do not just differ in average price ($331 vs. $165) the entire pricing relationship differs structurally between the two segments. Combined with the fact that over a third of listings in this dataset display no registration number, this suggests Toronto's short-term rental market is not one uniform market with a compliance overlay, but something closer to two distinct markets operating side by side. For Municipal Licensing and Standards, this is a useful data point for prioritizing enforcement: unlicensed activity is concentrated disproportionately at the lower end of the price spectrum.

The finding in Section 5.1 that unlicensed listings skew toward larger host portfolios, not smaller ones, complicates a simple "licensed = professional, unlicensed = casual" enforcement narrative and instead points toward Chapter 547's principal-residence rule mechanically sorting small-scale operators into the licensed pool and large-scale operators out of it.

9 Limitations

- **Single snapshot, not a time series.** All data reflects one Inside Airbnb scrape (May–June 2026); calendar.csv in this snapshot records availability but not per-night price, so no seasonal pricing analysis was possible.
- **licensed is a proxy for current compliance status, not a before/after comparison.** The Chow test compares listings that currently display a registration number against those that do not, rather than the same listings before and after the bylaw took effect.
- **Imputed, not observed, size variables.** bedrooms, beds, and bathrooms were imputed for a substantial share of listings (26–54% missing before imputation); results are conditional on the property-type-median imputation strategy used in Section 2.2.
- **Neighbourhood grouping is a modeling choice.** Collapsing 140 neighbourhoods to the top 14 + "Other" was necessary for a stable, interpretable dummy-variable block, but it does mean some finer-grained locational variation within the "Other" bucket is not captured.
- **No text mining.** The 576,000+ review comments in this Inside Airbnb export were not analyzed; only structured review counts, scores, and dates were used.
- **Occupancy proxy not confirmed bookings.** The Section 5.2 overpricing check uses calendar.csv's availability field as a proxy for occupancy; a blocked date can reflect an actual booking or simply a host choosing not to list availability, so the resulting correlation should be read directionally, not as a precise bookings model.
- **Correlational, not causal.** Every finding in this report, including the licensing result, describes an association the data supports, not a causal claim about what would happen if a specific listing's characteristics or licensing status changed.
- **Toronto only.** The model reflects Toronto's specific market and regulatory environment (Chapter 547) and is not assumed to generalize to other cities' short-term rental markets.

10 Appendix A: Full Regression Output

$R^2 = 0.303$, Adj. $R^2 = 0.301$, $F = 199.8$ ($p \approx 0$), $n = 12{,}003$, 26 predictors. Dependent variable: raw price ($).

Variable	Coef	Std Err	t	p
const	-39.016	43.021	-0.907	0.364
accommodates	22.794	2.640	8.633	0.000
bedrooms	36.916	6.103	6.049	0.000
beds	12.714	4.482	2.837	0.005
bathrooms	105.136	5.838	18.008	0.000
room: Hotel room	126.878	49.891	2.543	0.011
room: Private room	-64.244	7.531	-8.531	0.000
room: Shared room	-56.642	75.195	-0.753	0.451
nbhd: Bay Street Corridor	-10.927	22.340	-0.489	0.625
nbhd: Church-Yonge Corridor	-22.683	20.812	-1.090	0.276
nbhd: Dovercourt-Wallace Emerson-Junction	-105.068	22.612	-4.646	0.000
nbhd: Kensington-Chinatown	-72.051	22.573	-3.192	0.001
nbhd: Little Portugal	-46.955	24.924	-1.884	0.060
nbhd: Moss Park	-27.657	21.109	-1.310	0.190
nbhd: Niagara	21.238	19.771	1.074	0.283
nbhd: Other	-119.282	15.322	-7.785	0.000
nbhd: Palmerston-Little Italy	-44.864	26.045	-1.723	0.085
nbhd: South Riverdale	-117.851	24.806	-4.751	0.000
nbhd: Trinity-Bellwoods	-62.197	22.893	-2.717	0.007
nbhd: Waterfront Communities-The Island	-13.703	16.054	-0.854	0.393
nbhd: Willowdale East	-115.279	24.656	-4.676	0.000
nbhd: York University Heights	-119.069	27.975	-4.256	0.000
superhost	-24.550	5.576	-4.403	0.000
host tenure (years)	2.215	0.753	2.943	0.003
review score	19.727	8.155	2.419	0.016
minimum nights	-2.054	0.150	-13.690	0.000
licensed	76.672	6.928	11.067	0.000

Table A1. Baseline (raw-price) model — full coefficient table.

$R^2 = 0.730$, Adj. $R^2 = 0.729$, $F = 1244$ ($p \approx 0$), $n = 12{,}003$, 26 predictors. Dependent variable: log(price). OLS and heteroskedasticity-robust (HC1) standard errors and p-values are shown side by side; conclusions are unchanged under the robust specification.

Variable	Coef	SE (OLS)	p (OLS)	SE (HC1)	p (HC1)
const	4.590	0.067	0.000	0.075	0.000
accommodates	0.101	0.004	0.000	0.005	0.000
bedrooms	0.122	0.009	0.000	0.012	0.000

Variable	Coef	SE (OLS)	p (OLS)	SE (HC1)	p (HC1)
beds	-0.058	0.007	0.000	0.008	0.000
bathrooms	0.131	0.009	0.000	0.011	0.000
room: Hotel room	0.319	0.078	0.000	0.055	0.000
room: Private room	-0.581	0.012	0.000	0.012	0.000
room: Shared room	-0.698	0.117	0.000	0.121	0.000
nbhd: Bay Street Corridor	0.047	0.035	0.175	0.036	0.194
nbhd: Church-Yonge Corridor	-0.042	0.032	0.196	0.034	0.218
nbhd: Dovercourt-Wallace Emerson-Junction	-0.341	0.035	0.000	0.036	0.000
nbhd: Kensington-Chinatown	-0.159	0.035	0.000	0.035	0.000
nbhd: Little Portugal	-0.201	0.039	0.000	0.040	0.000
nbhd: Moss Park	-0.102	0.033	0.002	0.034	0.003
nbhd: Niagara	0.032	0.031	0.299	0.031	0.301
nbhd: Other	-0.425	0.024	0.000	0.025	0.000
nbhd: Palmerston-Little Italy	-0.150	0.041	0.000	0.047	0.001
nbhd: South Riverdale	-0.322	0.039	0.000	0.035	0.000
nbhd: Trinity-Bellwoods	-0.165	0.036	0.000	0.034	0.000
nbhd: Waterfront Communities-The Island	-0.012	0.025	0.627	0.025	0.634
nbhd: Willowdale East	-0.252	0.038	0.000	0.038	0.000
nbhd: York University Heights	-0.709	0.043	0.000	0.040	0.000
superhost	-0.045	0.009	0.000	0.009	0.000
host tenure (years)	0.006	0.001	0.000	0.001	0.000
review score	0.106	0.013	0.000	0.014	0.000
minimum nights	-0.015	0.000	0.000	0.001	0.000
licensed	0.316	0.011	0.000	0.017	0.000

Table A2. Log-price model — full coefficient table, OLS vs. robust standard errors.

11 Appendix B: Additional Diagnostics

B.1 Full-Model VIF

26 predictors, log-price model. All values remain under the conventional concern threshold of 10.

Variable	VIF
nbhd: Other	8.39
nbhd: Waterfront Communities-The Island	5.32
accommodates	4.59
bedrooms	4.23
beds	3.01
nbhd: Niagara	2.15
bathrooms	2.04
nbhd: Church-Yonge Corridor	1.94
nbhd: Moss Park	1.90
nbhd: Kensington-Chinatown	1.78
nbhd: Bay Street Corridor	1.73
room: Private room	1.73
nbhd: Dovercourt-Wallace Emerson-Junction	1.71
nbhd: Trinity-Bellwoods	1.68
nbhd: Willowdale East	1.54
licensed	1.54
nbhd: South Riverdale	1.53
nbhd: Little Portugal	1.51
minimum nights	1.48
nbhd: Palmerston-Little Italy	1.46
nbhd: York University Heights	1.44
host tenure (years)	1.18
superhost	1.09
review score	1.05
room: Hotel room	1.03
room: Shared room	1.01

Table B1. Full-model variance inflation factors.

B.2 Coefficient Stability: Full Sample vs. Excluding Cook's-D-Flagged Listings

Log-price model, full sample (n = 12,003) vs. training sample excluding the 623 listings flagged by Cook's Distance (Section 4.2). Sorted by absolute percentage change.

Variable	Coef (full)	Coef (excl. influential)	% change
nbhd: Waterfront Communities-The Island	-0.012	0.007	+155.4%
nbhd: Bay Street Corridor	0.047	0.019	-59.2%
nbhd: Niagara	0.032	0.048	+49.1%

Variable	Coef (full)	Coef (excl. influential)	% change
nbhd: Church-Yonge Corridor	-0.042	-0.056	-33.0%
review score	0.106	0.139	+31.4%
room: Hotel room	0.319	0.226	-29.3%
minimum nights	-0.015	-0.018	-22.1%
bedrooms	0.122	0.142	+16.2%
host tenure (years)	0.006	0.007	+15.4%
nbhd: Palmerston-Little Italy	-0.150	-0.173	-15.2%
beds	-0.058	-0.066	-14.9%
licensed	0.316	0.274	-13.4%
nbhd: Little Portugal	-0.201	-0.224	-11.4%
superhost	-0.045	-0.040	+10.7%
nbhd: Dovercourt-Wallace Emerson-Junction	-0.341	-0.306	+10.2%
nbhd: Willowdale East	-0.252	-0.228	+9.4%
bathrooms	0.131	0.120	-8.4%
nbhd: Kensington-Chinatown	-0.159	-0.171	-7.9%
nbhd: Other	-0.425	-0.400	+5.9%
accommodates	0.101	0.097	-4.6%
nbhd: Trinity-Bellwoods	-0.165	-0.158	+4.1%
room: Private room	-0.581	-0.599	-3.1%
nbhd: South Riverdale	-0.322	-0.314	+2.4%
const	4.590	4.484	-2.3%
nbhd: York University Heights	-0.709	-0.697	+1.6%
nbhd: Moss Park	-0.102	-0.103	-0.9%
room: Shared room	-0.698	-0.693	+0.7%

Table B2. Coefficient comparison, full sample vs. excluding influential listings.

12 Appendix C: Mathematical Formulation of the Hypothesis Tests

C.1 Joint F-Test on the Neighbourhood Block

Let β_nbhd,1, ..., β_nbhd,14 denote the coefficients on the 14 neighbourhood dummy variables in the log-price model (Section 3.4), estimated on the full sample (n = 15,003).

$$H_0: \beta_nbhd,1 = \beta_nbhd,2 = \ldots = \beta_nbhd,14 = 0 \quad vs. \quad H_1: \text{at least one } \beta_nbhd,k \neq 0$$

This was implemented as a Wald test with an explicit restriction matrix R (14 × 27): one row per neighbourhood dummy, a 1 in that dummy's column and 0s elsewhere, tested against $R\hat{\beta} = 0$. This is the general, robust way to test a block of coefficients and avoids any ambiguity in typing a hypothesis string by hand.

Result: F(14, 14,976) = 170.90, p ≈ 0.

C.2 Chow Test as an Interacted Regression

Let X denote the full set of 26 predictors used throughout this report, and let Licensed be the binary licensing indicator. The unrestricted (fully interacted) model is:

$$log(Price) = \beta_0 + \Sigma\beta k \cdot Xk + \gamma_0 \cdot Licensed + \Sigma\gamma k \cdot (Xk \times Licensed) + \varepsilon$$

$$H_0: \gamma_0 = \gamma_1 = \ldots = \gamma_{25} = 0 \quad \text{(one pooled pricing relationship for both groups)}$$

$$H_1: \text{at least one } \gamma \neq 0 \quad \text{(licensed and unlicensed listings price differently)}$$

The unrestricted model (51 predictors including interactions) was estimated on the full sample and achieves $R^2 = 0.738$, versus $R^2 = 0.729$ for the pooled model without interactions. A joint Wald test on the 26 restrictions (the Licensed main effect plus its 25 interaction terms) gives F(26, 14,951) = 62.07, p ≈ 0. H_0 is rejected decisively, meaning the entire pricing relationship, not merely its intercept, differs between licensed and unlicensed listings.

13 Appendix D: Data Cleaning and Imputation Detail

D.1 Row-Level Cleaning Steps

- price was converted from text (e.g. "$99.85") to numeric by stripping the currency symbol and thousands separators.
- neighbourhood_group_cleansed and instant_bookable were confirmed 100% missing for this Toronto snapshot (verified with an assertion before dropping, so the step fails loudly if a future snapshot populates either field) and dropped.
- bathrooms_text (e.g. "1 bath", "1.5 baths", "Half-bath") was parsed with a regular expression that extracts the leading numeric value, treating an unqualified "half-bath" as 0.5; this left only 0.2% missing versus 54.0% missing in the raw numeric bathrooms field.
- bedrooms, beds, and the parsed bathrooms field were imputed using the median value within each property_type category (e.g., "Entire condo", "Private room in home"), falling back to the overall sample median for any property type with no non-missing value for that field chosen over a single global median because a studio and a four-bedroom house are systematically different.
- review_scores_rating was imputed with the sample median (4.9), and a review_scores_missing flag (mean 0.179) was retained separately, since a listing with no reviews yet is systematically different from one with a mediocre rating.
- Rows with missing price (861, 5.4%) or missing minimum_nights (5) were dropped, since neither can be sensibly imputed for modeling purposes. Final modeling dataset: 15,003 listings (a loss of 866 rows, 5.5%, from the original 15,869).
- The licensed indicator was derived as license.notna(). 66.1% of listings carry a value in the license field, formatted as STR-YYYY-XXXXXX.
- The 14 most frequent values of neighbourhood_cleansed (out of 140) were kept as their own dummy category; all others were grouped into "Other," which remains the single largest category (7,621 listings) after grouping.

D.2 Train/Test Split

The cleaned, feature-engineered dataset (15,003 rows) was shuffled with a fixed random seed (NumPy default_rng(42)) and split 80/20: 12,003 listings for training (used for all model fitting, diagnostics, and the Section 5.2 occupancy check) and 3,000 held out for the Section 6 out-of-sample validation only. The fixed seed makes the split, and therefore every downstream number in this report, exactly reproducible.